

👤 Profile

Postdoctoral researcher and backend systems engineer with a strong track record of turning ideas into working systems, including database platforms, automated evaluation, and data infrastructure. Strong in Java, Python, SQL, object-oriented design, APIs, PostgreSQL, and end-to-end delivery from ambiguous problems to implementation, experimentation, and demo deployment.

🔧 Backend, Platform, and Data Systems Experience

Postdoctoral Research Fellow, The University of Melbourne, Australia Feb. 2024 – Present

Led and built end-to-end database systems, covering backend architecture, PostgreSQL services, evaluation harnesses, mentoring, and research-to-prototype delivery.

- Built backend-oriented data systems around PostgreSQL, including service APIs, Docker-based environments, extension hooks, workload generators, and reusable experiment pipelines. [Code 🔗]
- Proposed and prototyped a **metadata-native infrastructure** direction for agent workspaces, using metadata to support task-aware context routing, access control, and recoverable actions. [Code 🔗]
- Applied production-style engineering practices including modular architecture, Git workflows, automated testing, and CI/CD-style validation for repeatable experiments and demos. [Code 🔗]
- Developed a **layout advisory backend** for analytical data lake workloads, supporting workload ingestion, SQL parsing, and reusable decision logic for physical design. [Code 🔗] [VLDB 2026, submitted]
- Built **DriftBench**, a modular benchmarking and monitoring framework for evaluating system behaviour under workload and data drift, with configurable workload generation, repeatable experiments, and performance analysis. [Code 🔗] [VLDB 2026 🔗] [Website 🔗]
- Built **LLM-assisted query agents** that translate natural language into executable database workflows, including retrieval, query decomposition, and database-backed evaluation. [Code 🔗] [ADC 2025 🔗]

Data Scientist / Data Infrastructure Engineer, nftDb, Australia Feb. 2023 – Feb. 2024

Built data workflows, event-driven ingestion pipelines, and automation for NFT market data.

- Built event-driven data ingestion and automation pipelines for blockchain transaction data using Kafka and Airflow, including scheduling, monitoring, failure handling, and downstream analytics workflows.
- Developed a **RAG-based knowledge assistant** for documentation retrieval and engineering workflow support, combining retrieval, prompt design, and structured response generation.
- Used BigQuery to analyze large-scale transaction datasets, investigate user behaviour and marketplace activity, validate data quality, and improve reporting reliability.

Software Engineer (Java), Baidu, China

Jul. 2015 – Aug. 2017

Worked on Baidu's IM/chat backend platform for employees and business partners.

- Developed Java backend service components for Baidu's internal IM/chat platform, supporting messaging workflows, data access, API integration, debugging, deployment support, and production issue remediation.
- Designed messaging protocols and message deduplication mechanisms to improve delivery reliability, consistency, and fault tolerance across large-scale communication workflows.
- Improved database and service performance through profiling, query optimization, systematic tuning, and collaboration with cross-functional engineering teams.

⚙️ Technical Skills

Languages: Java, Python, SQL; working knowledge of C++ and Scala

Backend Engineering: REST APIs, backend services, object-oriented design, service integration, debugging

AI Systems: RAG, LLM-assisted workflows, tool orchestration

Data Systems: PostgreSQL, PostGIS, BigQuery, query optimization, indexing, benchmarking, vector databases

Infrastructure: Docker, Kafka, Airflow, Git, GitHub Actions, CI/CD, GCP, AWS, Apache Hudi

Frontend / Full-stack: React, Next.js, TypeScript, API integration, dashboard and demo development

🎓 Education

PhD in Computer Science, The University of Melbourne, Australia

2019 – 2023

M.S. in Computer Technology, Northeastern University, China

2013 – 2015

B.Eng. in Software Engineering, Northeastern University, China

2009 – 2013